Supplementary material for

From benchmark to pathway: a dynamic fuzzy soft set framework for circularity readiness and end-of-life photovoltaic strategy selection in the European Union

Muhammad Haris Saeed
Department of Chemistry, University of Management and Technology, Lahore 54700, Pakistan

S1. Dynamic fuzzy soft sets: operators used in the main text

Notation follows Saeed (2025) and Saeed et al. (2026). Let $\mathcal{S}_f$ and $\mathcal{G}_f$ be DFSSs over (U, E, T) with active sets A_t, B_t and memberships $\mu_{t,e}, \nu_{t,e}$.

Definition 1 (Dynamic fuzzy soft subset)**.** $\mathcal{S}_f \subseteq \mathcal{G}_f$ *iff for all* $t \in T$*:* $A_t \subseteq B_t$ *and* $\mu_{t,e}(u) \leq \nu_{t,e}(u)$ *for all* $e \in A_t$*,* $u \in U$*.*

Definition 2 (Union, intersection, complement)**.** $(\mathcal{S}_f \cup \mathcal{G}_f)(t) = (A_t \cup B_t,\ \max(\mu_{t,e}, \nu_{t,e}))$ *with missing memberships taken as* 0*;* $(\mathcal{S}_f \cap \mathcal{G}_f)(t) = (A_t \cap B_t,\ \min(\mu_{t,e}, \nu_{t,e}))$*;* $(\mathcal{S}_f)^c(t) = (A_t,\ 1 - \mu_{t,e})$*.*

Definition 3 (Time-slice and parametric reduction)**.** *For* $T' \subseteq T$*,* $\mathcal{S}_f|_{T'} = \{(t, A_t, F_t) : t \in T'\}$*; for relevance sets* $R_t \subseteq A_t$*,* $\mathcal{S}_f^R = \{(t, R_t, F_t|_{R_t})\}$*. Both commute with union and intersection.*

Reduction properties used in the paper. (i) If $A_t = A$ for all t, Eq. (2) of the main text reduces to a time-fuzzy soft set evaluation and, with equal weights and unit confidence, the Phase-3 score equals the pooled composite B1. (ii) Hence the divergence between DFSS and B1 vanishes when $|A_t|$ is constant and weights and confidence are uniform; with dimension-balanced weights and flag-based confidence the agreement is close but not exact, which is what Scenario S1 (near-constant cardinality; $\tau_b = 0.97$ against B1) shows empirically. (iii) The decomposition of Eq. (3) is exact by construction; its two terms are the parametric reduction of period t_1 to A_{t_0} and the residual.

Confidence-weighted local aggregation. Writing $\tilde{\mu}_{t,e}(u) = c_{t,e}(u)\mu_{t,e}(u)$ and $\tilde{w}_{t,e}(u) = w_{t,e}c_{t,e}(u)$, Eq. (2) is the ordinary weighted mean with observation-specific weights; it is monotone in μ, bounded in $[0, 1]$, and reduces to the DFSS Phase-2 mean when $c \equiv 1$.

Fuzzy arithmetic in Layer C. For triangular fuzzy numbers $\tilde{a} = (l, m, r)$: addition and positive scalar multiplication are componentwise; benefit normalisation divides by the maximum upper bound u^*; cost normalisation uses $(l^*/u,\ l^*/m,\ l^*/l)$ after shifting all values to be strictly positive; the graded mean integration representation is $(l + 4m + u)/6$.

S1.1 Supplementary figures

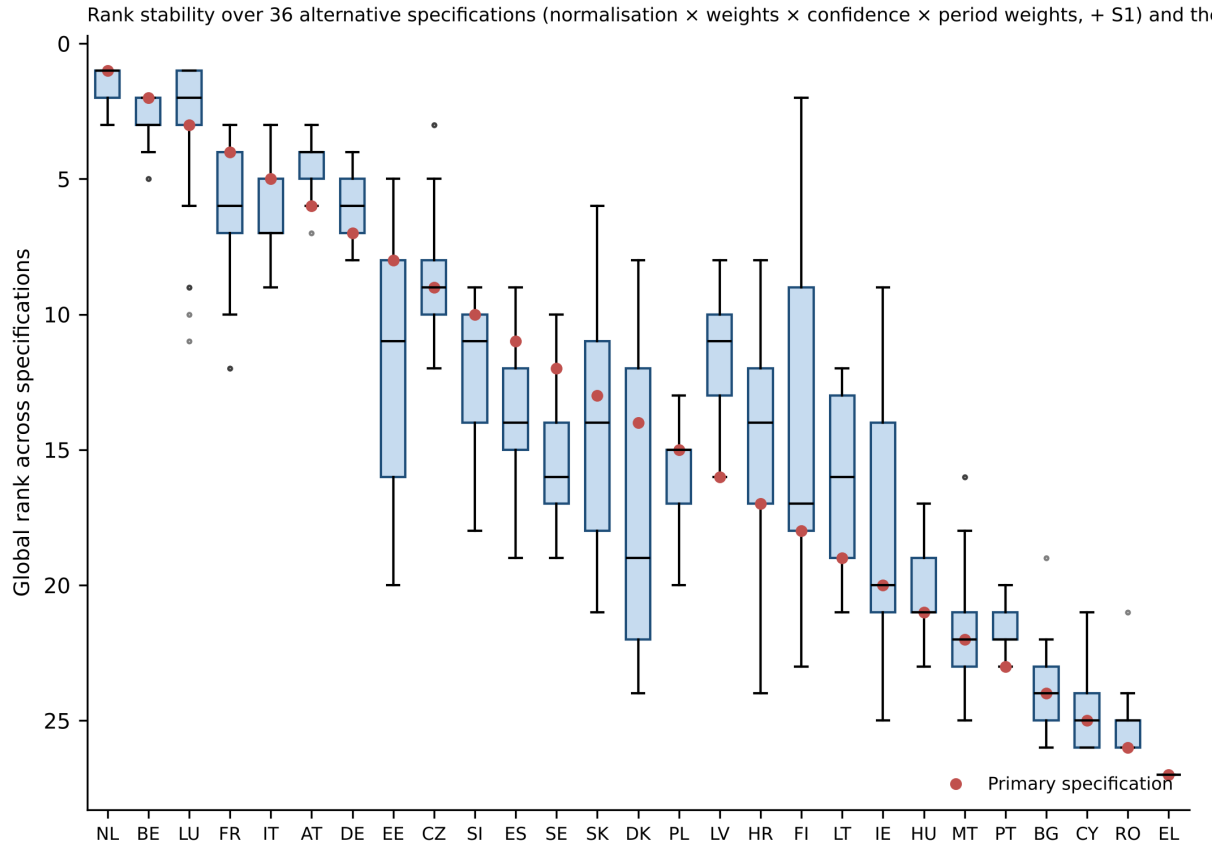


Figure S1. Global rank of each member state across the 36 alternative specifications of the sensitivity grid and Scenario S1 (boxes) and under the primary specification (red).

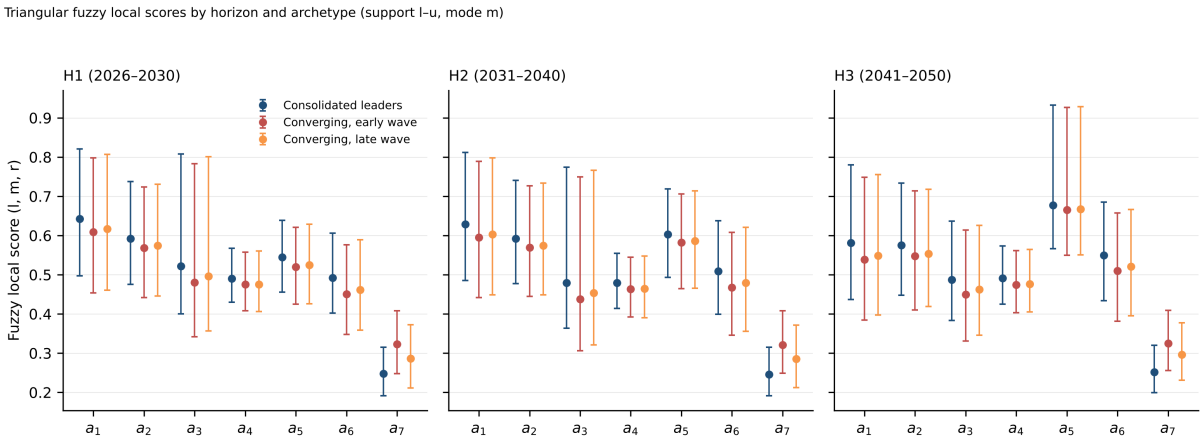


Figure S2. Triangular fuzzy local scores of the seven strategies a_1–a_7 by horizon and archetype (L, EW, LW; support l–r, mode m).

S2. Data, parameters and code

S2.1 Eurostat datasets

Circular economy indicators: cei_pc020, cei_pc030, cei_pc031, cei_pc032, cei_pc034, cei_pc035, cei_pc040, cei_pc050, cei_wm010, cei_wm011, cei_wm020 (W1501), cei_wm060, cei_srm030, cei_cie011 (PC_EMP_FTE), cei_cie012 (INV, GVA; PC_GDP), cei_cie020 (P_MHAB), cei_gsr010 (SWS, P_HAB), cei_gsr011, cei_gsr030. PV capacity: nrg_inf_epcrw (RA420, CAP_NET_ELC, MW). WEEE: env_waselee (EE_CPV_PVP, COL, T; ten-category scope of Annex I, reference years 2016–2018) and env_waseleeos (EE_LE_PVP, COL, T; six-category open scope of Annex III, 2019–2024); put-on-market tonnage from the same datasets (MKT). Population: tps00001. All retrieved through the dissemination API (JSON-stat 2.0) on 20 October 2026, including observation status flags.

S2.2 Active parameter sets

Scenario S2: $A_{2012} = A_{2016} = \{$WGDP, MW, RRMW, RRAW, RRPW, RRWEEE, CMU, INV, GVA, EMP, PAT$\}$; $A_{2020} = \{$WGDP, MW, RRMW, RRPW, RRWEEE, CMU, INV, GVA, EMP, PAT, FW$\}$; $A_{2023} = \{$MF, RP, WG, WGDP, MW, FW, PW, PPW, RRMW, RRPW, RRWEEE, CMU, INV, GVA, EMP, CF, GHG, MID$\}$. Scenario S1: all indicators reported by ≥ 22 member states in the window.

S2.3 Layer B parameters

Weibull $F(t) = 1 - \exp[-(t/\eta)^{\alpha}]$ with characteristic life $\eta = 30$ yr (IRENA/IEA-PVPS 2016 convention); shapes: early-loss 2.4928, regular-loss 5.3759. Calibrated early share $\pi = 0.18$ (pooled log-least-squares on 71 country-years with > 100 t collected, 2016–2024, Annex-I category EE_CPV_PVP for 2016–2018 and open-scope Annex-III category EE_LE_PVP for 2019–2024; 128 country-years with positive collection; HU-2018 excluded). Country-specific fits (three or more qualifying years): DE 0.05, CZ 0.03, BE 0.10, ES 0.00, IT 0.11, NL 0.14, PT 0.16, FR 0.28, PL 0.89. Mass check: Eurostat PV modules put on the market (t) versus capacity additions times vintage mass factor, 107 country-years, median ratio 0.97, IQR 0.72–1.30. Mass per MW by vintage: 140 (1990), 120 (2000), 100 (2010), 85 (2016), 65 (2020), 55 (2024), 42 (2050) t/MW, linearly interpolated (IRENA/IEA-PVPS 2016 exponential fit; current module data sheets). Composition (mass shares, IRENA/IEA-PVPS 2016): glass 0.755, polymers 0.10, aluminium 0.08, silicon 0.05, copper 0.010, silver 0.0005, other 0.0045. Interior capacity reporting gaps (DK 2014–2017, 2019; LU 2018–2020; LV 2017; RO 2023) linearly interpolated. Monte Carlo: 600 draws.

S2.4 Layer C literature-derived criterion values

Triangular fuzzy numbers (l, m, r) per alternative and horizon for C_5 net cost (EUR/t), C_6 quality-weighted recovery (%), C_7 avoided GHG (kg CO_2e/t), C_8 CRM recovery (%), C_9 employment (FTE/kt) are listed in the accompanying workbook `LayerC_results.xlsx`, sheet D6, and in `code/layer_c.py`. Anchors: D'Adamo et al. (2017) recycling cost 8–19 EUR/m^2; Markert et al. (2020) FRELP private end-of-life cost USD 6.7/m^2 and recovered-material value USD 13.6/m^2, with FRELP yields Al 96 %, glass 98 %, Cu 75 %, Si 39 %, Ag 11 % (Latunussa et al. 2016); Cucchiella et al. (2015); Deng et al. (2019); Heath et al. (2020); Tsanakas et al. (2020); Maani et al. (2020). Avoided-GHG values are relative ranges anchored on these LCAs and are subject to the ± 20 % perturbation analysis.

S2.5 Code

`eurostat.py` (API client), `build_panel.py` (panel and active sets), `layer_a.py`, `layer_b.py`, `layer_c.py` (models), `figures_a/b/c.py` and `make_tables.py` (outputs). Full pipeline: `python3 build_panel.py && python3 layer_a.py && python3 layer_b.py && python3 layer_c.py`. Python 3.12, pandas 3.0, scikit-learn 1.8, scipy 1.17.

S2.6 Institutional proxies

Scripts `layer_c_proxies.py` builds the country capacity indices (`results/C_country_capacity.csv`), archetype capacities and the fuzzy feasibilities (`results/C_institutional_proxy.csv`) from Eurostat env_waseleeos (EE6 and EE_LE_PVP), the CE monitoring framework panel and the World Bank WGI API (GOV_WGI_RQ.EST, GOV_WGI_RL.EST, 2021–2023 means). Raw and normalised indicator values for all 27 member states are in `results/C_proxies_raw.csv` and `C_proxies_norm.csv`.

S3. Requirement matrix and Delphi protocol

S3.1 Requirement matrix $\rho(a, c)$: rationale by alternative

a_1 **Repair and refurbishment** needs preparing-for-reuse standards and re-certification rules (C1 = 0.75), refurbishers and a used-module market (C3 = 0.75), and buyer acceptance of used modules (C4 = 0.75, rising to 1.00 in H3 when the vintages reaching end of life are old) (Tsanakas et al. 2020). Enforcement matters less (C2 = 0.50).

a_2 **Cascaded second use** is enabled by repowering rules and warranty transfer (C1 = 0.75) and by second-use demand (C3 = 0.50; C4 = 0.50 rising to 0.75 in H3).

a_3 **Export for second life** hinges on the used-equipment/waste boundary and shipment rules (C1 = 1.00) and on their enforcement (C2 = 1.00); it needs only traders (C3 = 0.25) and modest acceptance (C4 = 0.50).

a_4 **Bulk mechanical recycling** is already mandated by the WEEE Directive (C1 = 0.50, C2 = 0.50) and needs treatment capacity (C3 = 0.75); public acceptance is not binding (C4 = 0.25).

a_5 **High-value recycling** needs recovery targets or CRM crediting (C1 = 0.75) and, above all, recyclers and secondary-material buyers (C3 = 1.00 in H1, 0.75 in H2, 0.50 in H3 as the technology matures; Heath et al. 2020; Deng et al. 2019).

a_6 **Formalised take-back** needs extended-producer-responsibility (EPR) scheme design (C1 = 0.75), certified collection and leakage control (C2 = 0.75), installer and producer-responsibility-organisation (PRO) actors (C3 = 0.75, falling to 0.50 in H3 once networks exist) and willingness of owners to hand over modules (C4 = 0.75).

a_7 **Landfill/incineration** is the default where regulation, enforcement and participation are weak; strong institutions reduce its feasibility (inverse entries C1 = C2 = 1.00, C4 = 0.75); it needs little actor capacity (C3 = 0.25).

S3.2 Delphi protocol for expert validation

A two-round Delphi instrument was designed to elicit C1–C4 for every alternative, criterion and horizon on a five-level linguistic scale (VL, L, M, H, VH, mapped to the triangular fuzzy numbers $(0, 0, 0.25)$, $(0, 0.25, 0.5)$, $(0.25, 0.5, 0.75)$, $(0.5, 0.75, 1)$ and $(0.75, 1, 1)$ in `code/layer_c.py`) from a

panel of regulators, producer-responsibility organisations, installers, recyclers and researchers with experience in EU member states; Round 1 (20–30 minutes) also elicits criterion importance (1–9) and overall feasibility by member-state group; Round 2 presents the panel's aggregated ratings for revision. The instrument is provided as `Delphi_Round1_PV_EoL_questionnaire.docx` and as a Google Forms builder script; responses are ingested from `data/expert_ratings.csv` (columns: expert_id, horizon, archetype, alternative, criterion, rating) by `ingest_forms.py`, after which `layer_c.py` reports the Kendall agreement between proxy-based and expert-based rankings, a fuzzy consensus index and leave-one-expert-out stability. Recruitment for this study did not reach an adequate panel size before submission; the protocol is reported so that the proxy-based feasibilities can be tested.